"So any advanced semiconductor chip manufacturing, which is any of the leading nodes, I think the dependence on the world on one single large manufacturer from one part of the world is a real supply chain risk," he says.

The lesson is primitive but powerful. "We can't have one point of dependency for anything that we need," emphasises Viswanathan.

Without any hyperbole, he also breaks down how building semiconductors is intrinsically very hard.

"So it's a long journey. It's a complex sub supply chain," he explains. "It's sand and water. But even the sand and the water that is used is very different, it needs to be of certain accuracy or quality standards." And the fragility is real. "The smallest thing affects the fab, power affects the fab, vibrations affect the fab."

If you want a sense of scale, he zooms further in: "Now we're in the Angstrom era, where it's even smaller than nanometres now," calling it "one of the hardest things to do."

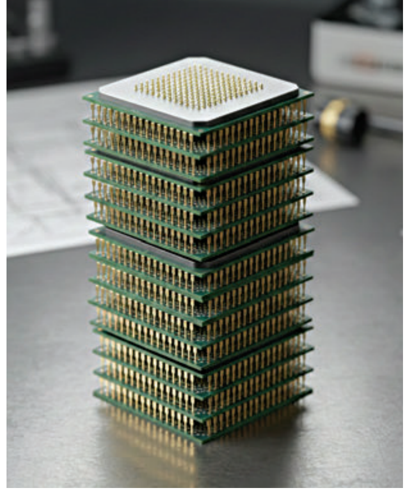
India's digital society needs lots of semiconductor chips

Here Viswanathan's argument turns distinctly Indian in how he frames chips as social infrastructure.

"We are one of those societies that is the most digital, right? Thanks to UPI, Aadhaar," he says. "Because we have IDs that are biometric, we have a payment system that is scalable. No other place has that."

Building your own chips is then fulfilling a foundational need, Viswanathan points out. "If we are a digital first society and the key to a digital infrastructure is a semiconductor chip, then having resiliency is the foundation of our society," he says. Because the alternative is unthinkable. "Like today, if I tell you there's no UPI, I will be lost, right? I don't have a plan B, right?"

That's the semiconductor argument in its simplest form. Not



prestige, not geopolitical theatre. Just plain and simple resilience.

For all the chest-thumping around "leading edge," Viswanathan prefers a gym analogy. "How do we start to build muscle? Think of it like your fitness journey," he says. "You can't get up one day and say I'm going to suddenly become extremely fit."

This is why he supports starting with technologies the ecosystem can realistically execute. "Let's get into the trailing nodes first. So 28 nanometers, the key trailing technologies."

When I ask whether India needs a leading-edge fab to become a semiconductor power, he doesn't hedge.

"I think that's the wrong question," says Viswanathan. Instead, he flips the causality. "Our race is not towards a leading edge fab," he says. "Our race should be that the market is consuming such leading edge services and technologies that by default, a leading edge fab becomes a market necessity."

That inversion matters. Build the market. As a consequence, let the fab become inevitable.

Demand generation is key

Ask what truly limits India, and Viswanathan goes straight to an inconvenient reality. "What's your biggest limiter? Demand." he says.

Restrictions can be negotiated. Talent exists. The constraint is market depth. Which is why he rejects the "backup supplier" narrative.

"We should not be a plus one option from a supply chain. We should be a plus one option from a market size and demand perspective."

His favourite illustration is the personal computer.

"We are the only country with less than 10% household PC penetration. Less than 10%," he says – before dismantling the affordability excuse: "a TV today is at 95%. Same cost as a PC. Right. A phone today is already at 95%. Same cost as a PC."

The conclusion is blunt. "We are not using compute the way it is meant to be used. With AI, for every kid, as much as a textbook and a notebook is important, a computer is an important necessity as well."

This is where the semiconductor story stops being about fabs and starts being about national capability design. A product nation mindset. Deep domestic consumption. Compute as habit, not aspiration. Otherwise, as he warns implicitly, design talent alone will not sustain strategic relevance for very long.

Viswanathan isn't anti-incentive. He just doesn't want dependency. "It shouldn't be a drug that it goes away, then you don't have anything to show." Ignition spark is fine, but addiction is not.

His operational fix is pragmatic. Create structured domestic demand – education is his example – and connect it directly to local manufacturing. Connect the dots and close the loop. Because in the end, markets decide.

"Eventually any innovation needs to be consumed somewhere." Otherwise, he adds with quiet finality, "it just becomes a good PPT slide."

India will miss a milestone here, hit one there, and argue loudly about both. The semiconductor journey will be expensive, technically brutal, and politically charged. Viswanathan's advice is stubbornly practical. "Keep persevering through the journey is what I would say." In an industry governed by physics, capital, and patience, that might be the only advice that truly scales. d